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Activity 3: SIEM Log Analysis & Incident Detection

PART 1: Quick Understanding Check

1. What is SIEM?

- SIEM stands for Security Information and Event Management. It is a cybersecurity system that collects, monitors, and analyzes logs from different devices and systems to detect possible security threats.

2. What is the role of log collection?

- Log collection gathers records of activities from servers, firewalls, applications, and devices. These logs help security teams monitor system activities and investigate suspicious behavior.

3. What is event correlation?

- Event correlation is the process of connecting related logs or events to identify possible attacks or security incidents. It helps analysts understand the full attack scenario instead of looking at single events only.

4. Difference between SIM and SEM

- SIM focuses on storing and managing logs for long-term analysis and reporting. SEM focuses on real-time monitoring and detecting security threats as they happen.

5. Why is SIEM important in cybersecurity?

- SIEM is important because it helps organizations quickly detect and respond to cyberattacks. It improves security monitoring and reduces the time needed to investigate incidents.

PART 2: Log Analysis

1. What suspicious activities can you identify?

- The identified red flags include numerous unsuccessful login entries, the presence of a questionable IP, detected malware, access to restricted data, and the lockout of an administrator account.

2. What type of attack is occurring?

- The incident seems to be a brute force attempt paired with a potential malware breach and illegal entry into the network.

3. Which logs are related?

- Relevant records involve server authentication logs, firewall traffic data, endpoint protection alerts, and logs from application usage.

4. What makes this behavior malicious?

- The recurring failed entry attempts, a dubious source IP, and unapproved data retrieval represent irregular patterns suggesting an adversary is attempting to compromise the environment.

5. What should be the immediate response?

- Initial countermeasures involve blacklisting the suspect IP, hardening the compromised account, performing a malware scan, and conducting a deep dive into all connected event logs.

PART 3: Event Correlation

1. Identify which events are connected

The following occurrences appear linked:

- E1 – Ten unsuccessful authentication attempts
- E2 – Dubious IP address activity
- E3 – Locking of administrative credentials
- E4 – Malicious software identification
- E5 – Unapproved retrieval of information

These records are interrelated as they demonstrate a chronological progression of questionable behaviors.

2. What is the possible attack scenario?

- A threat actor likely tried to breach the environment via repetitive login failures originating from a suspect source. Once the target account was restricted, malware was presumably deployed to facilitate entry and exfiltrate confidential data.

3. Assign a severity level

Severity Level: CRITICAL

4. Explain your reasoning

This situation is deemed critical due to the potential for illicit access, infection by malware, and the risk of data exfiltration. The involvement of multiple platforms significantly elevates the organizational security risk.

PART 4: SIEM Alert Creation

ALERT LEVEL: CRITICAL

INCIDENT NAME:

Brute Force Activity Involving Malware and Unapproved Access

DATE/TIME:

09:16 AM – 09:20 AM

DESCRIPTION:

A series of failed authentication attempts was identified from a questionable IP. Following these repetitions, the admin profile was locked. Subsequent malware discovery on the endpoint and unapproved data retrieval suggest a major infrastructure breach.

AFFECTED SYSTEMS:

- Identity Management Server
- Network Perimeter Firewall
- User Endpoint Equipment
- Database and Software Applications

EVIDENCE:

- Ten unsuccessful entry events
- Dubious source IP logs
- Locked administrator account status
- Endpoint malware detection alerts
- Logs indicating unapproved data retrieval

RECOMMENDED ACTION:

- Restrict the questionable source IP
- Modify passwords for impacted profiles
- Execute a comprehensive malware analysis
- Restore and update security software
- Analyze all event records thoroughly
- Continually observe for further anomalies

PART 5: Dashboard Interpretation

Scenario:

A SIEM dashboard shows:

- 120 failed login attempts
- 5 malware detections
- 3 suspicious IP addresses
- Most targeted system: Database Server

1. What is the biggest threat?

- The primary danger involves the likely unapproved entry into the database environment through authentication failures and malware exploitation.

2. Which system is at risk?

- The database server remains the most vulnerable asset given it is the central focus of the attack.

3. What should be prioritized?

- Immediate priority should be placed on database server protection, blacklisting suspect IPs, and investigating detected malware.

4. How does SIEM help in this situation?

- SIEM facilitates the gathering of records across platforms, identifying anomalies instantly, and empowering defensive teams to mitigate threats more effectively.

PART 6: Reflection Critical Thinking

1. What would happen without SIEM?

- Without SIEM, security teams may not detect attacks quickly. Important logs could be missed, making investigations slower and increasing the chance of system damage.

2. How does SIEM improve response time?

- SIEM improves response time by automatically monitoring logs and generating alerts when suspicious activities are detected. This allows security teams to react immediately.

3. Which SIEM tool would you like to use and why?

- I would like to use Splunk because it is widely used, user-friendly, and provides real-time monitoring and log analysis features that help detect threats effectively.